Sec: SR.IIT_*COSC(MODEL-A&APEX) | SPL GTM-12 | Date: 23-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**D**	2	**D**	3	**D**	4	**D**	5	**C**
6	**C**	7	**D**	8	**D**	9	**D**	10	**A**
11	**B**	12	**B**	13	**B**	14	**D**	15	**A**
16	**D**	17	**A**	18	**B**	19	**C**	20	**B**
21	**50**	22	**128**	23	**15**	24	**3**	25	**15**

CHEMISTRY

26	**A**	27	**C**	28	**D**	29	**C**	30	**B**
31	**B**	32	**A**	33	**A**	34	**A**	35	**A**
36	**A**	37	**B**	38	**A**	39	**C**	40	**A**
41	**D**	42	**A**	43	**A**	44	**D**	45	**A**
46	**50**	47	**827**	48	**4**	49	**5**	50	**96**

MATHEMATICS

51	**D**	52	**A**	53	**B**	54	**A**	55	**D**
56	**B**	57	**C**	58	**A**	59	**B**	60	**A**
61	**C**	62	**A**	63	**C**	64	**C**	65	**A**
66	**C**	67	**C**	68	**B**	69	**B**	70	**B**
71	**1**	72	**17**	73	**89**	74		75	

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SOLUTIONS

PHYSICS

1. $\frac{dE}{E} = 2\frac{dA}{A} + \mathrm{A}\frac{dt}{t} \times t$
2. Conceptual
3. Conceptual
4. Conceptual
5. $T = \frac{hr\rho g}{2\cos\theta}$
6. $\lambda = \frac{1}{\sqrt{2}\pi d^2 n} = V = f\lambda$
7. $E = -\frac{dv}{dx}$
8. Conceptual
9. $U = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 - V_2)^2$
10. $i = \frac{E}{2R}$
11. Conceptual
12. $a\sin\theta = n\lambda$
13. Conceptual
14. Conceptual
15. Conceptual
16. $\tau = I\alpha$
17. APPLY LCM
18. $W = U_f - U_i$

    $U = \frac{GMm_0}{\sqrt{R^2 + x^2}}$
19. $T = 2\pi\sqrt{\frac{m}{k}}$
20. $dQ = dU + dW$
21. $f = \frac{v}{4l} : l = \frac{\lambda}{4}$
22. $V_B = n^{2/3}v_5$
23. Conceptual
24. Conceptual
25. Conceptual

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CHEMISTRY

26. Total weight of H_2SO_4 = (150 × 80/100) + (250 × 20/100) = 120 + 50 = 170 g

Total weight of solution = 150 + 250 = 400 g

Total weight of H_2O = 400 - 170 = 230 g

Moles of H_2SO_4 = 170 / 98 = 1.735 mol

Moles of H_2O = 230 / 18 = 12.778 mol

Mole fraction of H_2SO_4 = 1.735 / (1.735 + 12.778) = 1.735 / 14.513 = 0.1196

27. Step 1: But-2-yne (CH_3-C≡C-CH_3) on hydration with Hg^{+2} / H_2SO_4 gives a mixture of ketones. Following Markownikoff's rule, the product is CH_3-CO-CH_2-CH_3 (butan-2-one). A = Butan-2-one.

Step 2: Butan-2-one reacts with CH_3MgBr. The nucleophile (CH_3^-) attacks the carbonyl carbon, followed by protonation to give a tertiary alcohol. B = CH_3-C(OH)(CH_3)-CH_2-CH_3 (2-methylbutan-2-ol)

Step 3: Dehydration with Conc. H_2SO_4 gives the more substituted alkene (Saytzeff product). C = CH_3-C(CH_3)=CH-CH_3 (2-methylbut-2-ene)

Step 4: Ozonolysis of 2-methylbut-2-ene [$(CH_3)_2C$=CH-CH_3] with reductive work-up gives: $(CH_3)_2C$=O (propanone) and CH_3-CH=O (ethanal) D = Propanone, E = Ethanal

28. Moles of metal M = 4.8 / 24 = 0.2 mol

Moles of O atoms = 3.2 / 16 = 0.2 mol

Mole ratio M : O = 0.2 : 0.2 = 1 : 1

So the formula is MO.

(A) Formula is MO. So A is true.

(B) In MO, oxidation state of M is +2. Equivalent weight = Atomic mass / n-factor = 24 / 2 = 12 g/eq. B is true.

(C) For 6.0 g metal, moles = 6 / 24 = 0.25 mol. From formula MO, 1 mol M reacts with 1 mol O atoms = 0.5 mol O_2 molecules. So 0.25 mol M requires 0.125 mol O_2. Mass of O_2 = 0.125 × 32 = 4.0 g. C is true.

(D) For 4.8 g metal (0.2 mol), moles O_2 required = 0.1 mol. Volume at STP = 0.1 × 22.4 = 2.24 L. D is true.

29. Electronic configurations:

K (Z=19): [Ar] $4s^1$

Ca (Z=20): [Ar] $4s^2$

IE_1 (K) = 419 kJ/mol, IE_1 (Ca) = 590 kJ/mol. So IE_1 (K) < IE_1 (Ca). Statement-I is true.

IE_2 (K): K^+ ([Ar]) → K^{2+} ([Ar] $3p^5$). This requires removing an electron from a stable noble gas configuration, so IE_2 is very high (3052 kJ/mol).

30. Step 1: Benzene undergoes Friedel-Crafts acylation with CH_3COCl in presence of $AlCl_3$ to give acetophenone ($C_6H_6COCH_3$). A = Acetophenone.

Step 2: Acetophenone undergoes Clemmensen reduction (Zn(Hg), HCl, Δ) which reduces the carbonyl group to methylene, giving ethylbenzene ($C_6H_5-CH_2-CH_3$). B = Ethylbenzene.

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31. Volume of N_2 gas collected, V = 40 mL = 0.040 L

Temperature, T = 300 K

Total pressure = 740 mm Hg. Aqueous tension = 24 mm Hg.

Pressure of dry N_2 gas, P = 740 - 24 = 716 mm Hg = 716/760 atm.

Using PV = nRT to find moles of N_2:

n = (P × V) / (R × T) = (716/760 × 0.040) / (0.0821 × 300)

= (0.9421 × 0.040) / (24.63) = 0.03768 / 24.63 = 0.00153 mol

Mass of N_2 = n × 28 = 0.00153 × 28 = 0.04284 g

Mass of nitrogen in compound = mass of N_2 = 0.04284 g

Percentage of N = (0.04284 / 0.25) × 100 = 17.14%

32. For zero-order, rate = k, so units of k = units of rate $\left(mol\ L^{-1}s^{-1}\right)$. True.

(B) For first-order, $t_{1/2} = \ln 2 / k$, independent of $[A]_o$. True.

(C) Catalyst lowers activation energy. True.

(D) Rate constant increases with temperature (Arrhenius equation). False.

So A, B, and C are correct.

33. The parent chain must include the aldehyde carbon (C-1). Identify the longest continuous carbon chain including the -CHO group.

Structure: $CHO - CH_2 - CH(OH) - CH(CH_3) - CH_2 - CH_3$

34. Statement-I: Acidity of hydrides increases down the group due to decrease in bond dissociation enthalpy and increase in size of the anion, making it easier to release H^+.

So $H_2O < H_2S < H_2Se < H_2Te$ is correct. True. Statement-II: Thermal stability decreases down the group because bond dissociation enthalpy decreases as the size of the central atom increases.

So $H_2O > H_2S > H_2Se > H_2Te$ is correct. True. Both statements are true.

35. A. Glucose with Tollen's reagent (ammoniacal $AgNO_3$) oxidizes the aldehyde group to carboxylic acid, giving gluconic acid. A-III.

B. Glucose with excess acetic anhydride acetylates all five -OH groups, giving glucose pentaacetate. B-IV.

C. Glucose with hydroxylamine reacts with the aldehyde group to form glucose oxime. C-I.

D. Glucose with conc. HNO_3 oxidizes both the aldehyde group and the terminal $-CH_2OH$ group to carboxylic acids, giving saccharic acid (a dicarboxylic acid). D-II.

So correct match is A-III, B-IV, C-I, D-II.

36. Dipole moment values (approx. in Debye):

- NH_3: 1.47 D (N has 1 lone pair)
- PH_3: 0.58 D (P has 1 lone pair)
- AsH_3: 0.20 D (As has 1 lone pair)
- SbH_3: 0.12 D (Sb has 1 lone pair)
- H_2O: 1.85 D (O has 2 lone pairs)

The molecule with the highest dipole moment is H_2O (1.85 D). Its central atom (O) has 2 lone pairs of electrons.

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37. Statement-I: IE_1 values (kJ/mol):
Li (520), Be (899), B (801), C (1086)
Order: Li (520) < B (801) < Be (899) < C (1086).
Statement-I is correct.
Statement-II: Electron gain enthalpy values (kJ/mol, negative values indicate energy released):
F: -328, Cl: -349, Br: -325, I: -295
Order of magnitude (most negative to least): Cl (-349) > F (-328) > Br (-325) > I (-295).
Statement-II is correct.
Both statements are true.

38. For a basic buffer, pOH = pKb + log([Salt] / [Base])
pH = 14 - pOH = 14 - [pKb + log([Salt] / [Base])] = 14 - 4.74 - log([Salt] / [Base]) = 9.26 - log([Salt] / [Base])
We need pH = 9.26, so log([Salt] / [Base]) = 0, which means [Salt] = [Base] in the final solution.
This implies millimoles of salt = millimoles of base in the buffer.

39. For a hydrogen-like ion with atomic number Z, the energy of a transition is:
$\Delta E = 13.6 \times Z^2 \times (1/n_1^2 - 1/n_2^2)$ Ev

40. Characteristics of a secondary standard:
• Less pure than primary standards - A is correct.
• Their solutions must be standardized against a primary standard - B is correct.
• They may or may not be hygroscopic - C is false.
• High molar mass is a characteristic of primary standards, not secondary - D is false.
• They cannot be used directly to prepare standard solutions of accurate concentration (they need standardization) - E is false.
So only A and B are correct.

41. $\left[Ni(CN)_4\right]^{-2}$: Ni is in +2 oxidation state. Ni (Z=28):
[Ar] $4s^2 3d^8$. Ni^{2+}: [Ar] $3d^8$.
CN^- is a strong field ligand, so it causes pairing of electrons. The complex is diamagnetic, so all electrons are paired.
Electronic configuration: $3d^8$ with strong field ligand → all 8 electrons paired in d orbitals
($3d^8$: $t_2g^6 eg^2$ with pairing, actually in square planar, all paired).
Hybridization: dsp^2 (square planar).
(A) Square planar complexes of type $\left[Ma_2b_2\right]$ can show geometrical isomerism. $\left[Ni(CN)_4\right]^{-2}$ has all four ligands same, so no geometrical isomerism. A is incorrect.
(B) $\left[Ni(CN)_4\right]^{-2}$ is known to be yellow-orange in color. B might be correct.
(C) Diamagnetic means μ = 0 BM. C is correct.
(D) Hybridization is dsp^2, not sp^3. D is incorrect.
(E) dsp^2 is inner orbital complex (uses (n-1)d orbitals), not outer orbital. E is incorrect.

42. Reducing power is the tendency to lose electrons (get oxidized). It is inversely related to the standard reduction potential (SRP). The more negative the SRP, the stronger the reducing agent.
Order of SRP: Zn^{2+}/Zn (-0.76) < Fe^{2+}/Fe (-0.44) < Cu^{2+}/Cu (+0.34) < Ag^+/Ag (+0.80). (More negative to more positive).
So reducing power order: Zn > Fe > Cu > Ag.

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43. A has molecular formula C_8H_8O. It gives a positive Fehling's test (reddish-brown precipitate of Cu_2O), so it must be an aldehyde. Options (1), (2), and (3) are aldehydes. Option (4) is a ketone, so it won't give Fehling's test.
It undergoes cross-aldol condensation with acetophenone to give a single product. For a single product, A must have α-hydrogens. All three aldehydes have α-hydrogens. But we need to check which one gives a single product with acetophenone.
Phenylacetaldehyde ($C_6H_5-CH_2-CHO$) has ∝ - hydrogens on the CH_2 group. With acetophenone, it will give a single aldol product. 4-Methylbenzaldehyde ($CH_3-C_6H_6-CHO$) has no ∝ - hydrogens on the aldehyde carbon (it's an aromatic aldehyde). It cannot undergo aldol condensation as it has no ∝ - H. So it won't react with acetophenone via aldol.
2-Methylbenzaldehyde also has no ∝ - H on the aldehyde carbon. So it won't react.

44. 2-Methylbutan-2-ol is $(CH_3)_2C(OH)-CH_2-CH_3$, a tertiary alcohol.

46. $$E_{cell} = E^o_{Ag/Ag^+} + \frac{0.06}{1} \times log \frac{[Br]}{K_{sp}(AgBr)} + E^o_{Ag^+/Ag} + \frac{0.06}{1} \times log \frac{K_{sp}(AgCl)}{[Cl^-]}$$

$$0.12 = 0.06\ log \frac{[Br^-]}{[Cl^-]} \times \frac{K_{sp}(AgCl)}{Ksp(AgBr)}$$

$$2 = \log \frac{1}{0.5} \times \frac{K_{sp}(AgCl)}{Ksp(AgBr)}$$

$$\frac{K_{sp}(AgCl)}{Ksp(AgBr)} = \frac{100}{2} = 50$$

47. $$K(s) + \frac{1}{2}F_2(g) \xrightarrow[\frac{1}{2}\Delta H_a]{\Delta H_a} K(g) + F(g)$$

$\downarrow \Delta H_f$ $\downarrow IE+EA$

$$K_f(s) \xleftarrow{\Delta H^o lattice} K^+_{(g)} + F^-_{(g)}$$

Born Haber Cycle
Heat of formation - Heat of Atomization - Sum of I.E - Sum of E.A = Lattice Energy
= -567 - (90 + 1/2 × 158) - 419 - 328 = Lattice Energy
= -567 - (90 + 79) - 419 - 328 Lattice energy
Lattice energy = -1483 kJ
Now, it is Exothermic Reaction therefore we will put (-) sign before Lattice Energy.
∴ -(-1483) = +1483 kJ

48. V_2O_5 — Amphoteric
Cr_2O_3 → Amphoteric
$Mn_2O_7^+$ — Acidic
V_2O_3 — Basic
V_2O_4 — Basic

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x = 1

Primary valency = charge on metal ion

= +3

y = 3

(x + y) = 4

49. Reaction: Aniline ($C_6H_6NH_2$) + Acetic anhydride → Acetanilide ($C_6H_5NHCOCH_3$) + CH_3COOH

Molar mass of Aniline ($C_6H_5NH_2$) = (6×12) + (7×1) + 14 = 72 + 7 + 14 = 93 g/mol

Molar mass of Acetanilide (C_8H_9NO) = (8×12) + (9×1) + 14 + 16 = 96 + 9 + 14 + 16 = 135 g/mol

Moles of aniline taken = 4.65 / 93 = 0.05 mol Theoretical mass of acetanilide = 0.05 × 135 = 6.75 g

Actual mass (80% yield) = 6.75 × 0.80 = 5.4 g Nearest integer = 5

50. 96 From the Arrhenius equation in logarithmic form:

$$log\ k\ =\ log\ A\ -\ \frac{Ea}{2.303RT}$$

MATHS

51. $A=\begin{pmatrix}1 & -n & 1\\ 1 & n-3 & n+2\\ 0 & n-2 & 1\end{pmatrix}$

$\det(A)=-n^2+3n-1$

The possible values of n are $\{1,2,3,4,5,6\}$

The sum is $k+(1+2+3+4+5+6)\ Sum=6+21=27$

52. $M=\frac{250k+251k}{2}=\frac{501k}{2}=250.5k$

$$MD=\frac{1}{n}\sum_{i=1}^{n}|x_i-M|$$

$$MD=\frac{1}{500}\sum_{i=1}^{500}|ik-250.5k|$$

$$MD=\frac{|k|}{500}\sum_{i=1}^{500}|i-250.5|$$

The sum can be calculated as

$$\sum_{i=1}^{500}|i-250.5|=(|1-250.5|+\ldots+|250-$$

$$=(249.5+\ldots+0.5)+(0.5+\ldots+249.5)$$

$$=2\times(0.5+1.5+\ldots+249.5)$$

$$\sum_{i=1}^{500}|i-250.5|=2\times0.5\times250^2=250^2$$

53. If $y=0, x^2\le40\Rightarrow x\in\{0,\pm1,....,\pm6\}(13\ values)$

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If $y = \pm 1, x^2 \leq 40 - 9 = 31 \Rightarrow x \in$ $\{0, \pm 1, \ldots, \pm 5\}$ ($2 \times 11 = 22$ values).

If $y = \pm 2, x^2 \leq 40 - 36 = 4 \Rightarrow x \in \{0, \pm 1, \pm 2\}$ $2 \times 5 = 10$ values).

If $|y| \geq 3, 9y^2 \geq 81$, which is > 40.

Total elements : 13 +22 +10 = 45

54. 1. Let z=x+iy. Substitute into equation : $2(x+iy)^2 + (x-iy) = 0$

2. Expand and separate real and imaginary parts: $2(x^2 - y^2) + x = 0$ and $y(4x - 1) = 0$.

3. **Case 1 ($y = 0$):** $2x^2 + x = 0 \Rightarrow x = 0, -1/2$. Roots: $z_1 = 0, z_2 = -1/2$. $|z|^2$ values: $0, 1/4$.

4. **Case 2 ($x = 1/4$):** $2(1/16 - y^2) + 1/4 = 0 \Rightarrow$ $1/8 - 2y^2 + 1/4 = 0 \Rightarrow 2y^2 = 3/8 \Rightarrow y^2 = 3/16$. $|z|^2 = x^2 + y^2 = 1/16 + 3/16 = 4/16 = 1/4$.

Total Sum: $0 + 1/4 + 1/4 + 1/4 = 3/4$.

55. Expand $\sin x \approx x - \frac{x^3}{6}$ and $\tan x \approx x + \frac{x^3}{3}$

Numerator: $a(x - \frac{x^3}{6}) + b(x + \frac{x^3}{3}) + cx^2 = (a + b)x + cx^2 + (\frac{b}{3} - \frac{a}{6})x^3$.

For the limit to exist and be non-zero, the coefficients of x and x^2 must be zero.

$a + b = 0$ and $c = 0$.

Limit value: $\frac{b}{3} - \frac{a}{6} = 2$.

Solving: Substituting $b = -a$, we get $-\frac{a}{3} - \frac{a}{6} = 2 \Rightarrow$ $-\frac{3a}{6} = 2 \Rightarrow a = -4$ and $b = 4$.

Final Answer: $a + b + c = -4 + 4 + 0 = 0$.

56. Separate variables $\frac{dy}{1+y^2} = \frac{dx}{2\sqrt{x}(1+x)}$

Integrate: Let $\sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}}dx = dt$. The RHS becomes $\int \frac{dt}{1+t^2}$.

Solve: $\tan^{-1}(y) = \tan^{-1}(\sqrt{x}) + C$.

Apply Condition: $\tan^{-1}(0) = \tan^{-1}(1) + C \Rightarrow$ $C = -\frac{\pi}{4}$.

Find Value: $\tan^{-1}(y(3)) = \tan^{-1}(\sqrt{3}) - \frac{\pi}{4} = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$.

Final answer : $y(3) = \tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}$

57. Logic for (S1): 1. Use the property that the slope of AO $\perp$ slope of BC.

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$m_{AO} = \frac{3}{2} \Rightarrow m_{BC} = -\frac{2}{3}$.

Form the equation of line BC using point B(4, -1) and verify if vertex C satisfies it.

58. Direct magnitude: $|c| = |a||b|\sin(\theta) = 2 \times 5 \times \sin(30^0) = 5$

Cauchy-Schwarz: $(c \cdot d)^2 \leq |c|^2|d|^2$.

Result: Since $|d| = 1$, the maximum value is $5^2 = 25$.

59. Logic: Perform the matrix multiplication (adj A)B.

Calculation:

$x' = (1)(4) + (-1)(2) + (1)(2) = 4$

$y' = (2)(4) + (0)(2) + (1)(2) = 10$

$z' = (-1)(4) + (2)(2) + (1)(2) = 2$

Result X= $\frac{1}{-2}[4, 10, 2]^T = [-2, -5, -1]^T$. Sum $= -8$.

60. Find the intersection point M (1,2,3). Use the distance formula d= $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} = 6$ for points $(1 + r, 2 + 2r, 3 + 2r)$ on L_1.

Result: r = ±2, leading to points (3, 6, 7) and (-1,-2,-1). Sum=16+(-4)=12

61. Logic: Since $e = \sqrt{2}$, then a = b. Substitute (6,3) into $\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$ find $a^2 = 27$

Calculation: Focal distance SS' = $2ae = 2\sqrt{27}\sqrt{2} = 6\sqrt{6}$ The height of the triangle is the y-coordinate of P, which is 3.

Result: Area $= \frac{1}{2} \times 6\sqrt{6} \times 3 = 9\sqrt{6}$ (Note: Adjusting constants for practice).

62. Logic: Find the intersection points of $x^2 + 3z - 4 = 0 \Rightarrow (x+4)(x-1) = 0$. Since $y^2 = 3x$. x must be 1

Integration: Total Area $= 2[\int_0^1 \sqrt{3x}\, dx + \int_1^2 \sqrt{4 - x^2}\, dx]$

63. Logic: Use the identity $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

Calculation: $4 \leq 100 - 4\lambda \leq 36$.

Solving: $-96 \leq -4\lambda \leq -64 \Rightarrow 16 \leq \lambda \leq 24$.

Result: Sum of integers from 16 to 24.

64. Logarithm: $x^2 - 5x + 6 > 0 \Rightarrow (x-2)(x-3) > 0 \Rightarrow x \in (-\infty, 2) \cup (3, \infty)$

$\sin^{-1}$ term: $-1 \leq \frac{|2x-1|}{5} \leq 1$. Since absolute value is always ≥ 0, we solve $|2x - 1| \leq 5$.

$-5 \leq 2x - 1 \leq 5 \Rightarrow -2 \leq x \leq 3$.

Intersection: $(-\infty, 2) \cup (3, \infty)$ intersected with $[-2, 3]$ gives $[-2, 2)$

65. Conceptual

66. Use the property PS = a - ex and PS' = a + ex.

$SP \cdot S'P = a^2 - e^2x^2 = 20$.

Find $e^2 = 1 - \frac{16}{25} = \frac{9}{25}$.

Solve for $x^2 = \alpha^2$ and then use the ellipse equation to find $y^2 = \beta^2$

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67. Curve S for $y^2 = 4ax$ is $y^2 = 2ax$

Let $Q = (x, y)$ and $P = (x_1, y_1)$.

$x = \frac{1 \cdot x_1 + 3 \cdot 0}{4} \Rightarrow x_1 = 4x$ and $y_1 = 4y$.

Substitute into S: $(4y)^2 = 2a(4x) \Rightarrow 16y^2 = 8ax \Rightarrow 2y^2 = ax$.

68. Logic: Solve $[x]^2 - 2[x] - 3 = 0 \Rightarrow ([x] - 3)([x] + 0) = 0$

Analysis: This is satisfied if [x] = 3 (which means 3 <x < 4) or [x] =- 1 (which means -1 < x <0)

Result: Since any x in these intervals works, there are infinitely many solutions.

69. Logic: $f(x+y) = f(x)f(y)$ implies $f(x) = a^x$ Given $f(1) = 3 f(x) = 3^x$

$g(x+y) = g(xy)$, $g(x) = 2$

Summation: The problem then involves solving a sum that combines a G.P. and an A.P. denominator.

70. Logic: Using the identity $\frac{C_r}{r+1} = \frac{^{n+1}C_{r+1}}{n+1}$, we find $P_n = \frac{1}{n+1} \sum_{r=0}^{n} (-1)^r \, ^{n+1}C_{r+1}$.

Simplification: $P_n = \frac{1}{n+1}[1 - (1-1)^{n+1}] = \frac{1}{n+1}$.

Summation: $\sum_{n=1}^{50}(n+1) = 2 + 3 + \cdots + 51 =$

$\frac{50}{2}(2 + 51) = 25 \times 53 = 1325$.

71. Step 1: Acute Angle Condition

$a \cdot b > 0$.

$(1)(2) + (\sqrt{2})(\sqrt{2}) + (\lambda)(-\lambda^2) > 0 \Rightarrow 4 - \lambda^3 > 0$

$0 \Rightarrow \lambda^3 < 4$.

Since λ is a positive integer, the only possible value is $\lambda = 1$

Step 2: Y-axis Direction Cosine

$\cos\phi = \frac{a_y}{|a|} = \frac{\sqrt{2}}{\sqrt{1^2+2+\lambda^2}} = \frac{\sqrt{2}}{\sqrt{3+\lambda^2}}$.

If $\lambda = 1, \cos\phi = \frac{\sqrt{2}}{\sqrt{4}} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$.

Since $\frac{1}{\sqrt{2}} > \frac{1}{2}$. $\lambda = 1$ is a valid integer solution. Result: 1 integer value.

72. The function $\left[x^{1/2}\right]$ changes at perfect squares:1,4,9,16.

$\int_0^1 0\,dx + \int_1^4 1\,dx + \int_4^9 2\,dx + \int_9^{16} 3\,dx$

$0 + (4-1) + 2(9-4) + 3(16-9) = 3 + 10 +$

$21 = 34$.

So, $\alpha = 34$.

$E = \frac{1}{\pi}[34 \cdot \frac{\pi}{9}] = 17$.

73. $p + q = 33 + 56 = 89$

74. step 1: Find b

$3b = 3 \Rightarrow b = 1$.

Let $a = 1 - d$ and $c = 1 + d$.

Step2 : Apply G.P condition

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$(3b^2)^2 = (a^2)(c^2) \Rightarrow 9 = (1-d^2)^2$

$1 - d^2 = -3 \Rightarrow d^2 = 4 \Rightarrow d = 2.$

Step 3 : Calculate

$a = -1, b = 1, c = 3.$

$a^2 + b^2 + c^2 = 1 + 1 + 9 = 11$

Result 11

75. Logic we need to list cases for the smallest element $x \in \{2,4,6,8,10\}$ Ans 256

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